$$y = x^2(n) \rightarrow$$

$$a_1 T[x_1(n)] = a_1 x_1^2(n)$$

$$a_2 T[x_2(n)] = a_2 x_2^2(n)$$

$$a_1 x_1^2(n) + a_2 x_2^2(n)$$

$$x_3(n) = a_1 x_1(n) + a_2 x_2(n)$$

$$T[x_3(n)] = ? = T[a_1 x_1(n) + a_2 x_2(n)]$$

$$= (a_1 x_1(n) + a_2 x_2(n))^2$$

x_3

$$= a_1^2 x_1^2(n) + a_2^2 x_2^2(n) + 2a_1 a_2 x_1(n) x_2$$

$$x(n) = \sum_{k=-\infty}^{+\infty} x(k)\ \delta(n-k)$$

$$x(n) = \{ \overset{n=-1}{2}, \underset{\uparrow}{\overset{n=0}{4}}, \overset{n=1}{0}, \overset{n=2}{3} \}$$

Express $x(n)$ into a sum of weighted impulse sequences

$\hookrightarrow \delta(n)$

at what values of n, $x(n)$ is NON-ZERO?

$n = -1, 0, 2$

$$x(n) = 2\,\delta(n+1) + 4\,\delta(n) + 3\,\delta(n-2)$$

$$h(n) \equiv T[\delta(n)]$$

impulse response of a system

is the response of the system to the $\delta(n)$ signal

$$y(n) = T[x(n)] = \sum_{k=-\infty}^{\infty} x(k)\, \underline{\underline{h}}(n-k)$$

n_0

$$y(n_0) = \sum_{k=-\infty}^{+\infty} x(k)\, h(n_0-k)$$

Convolution Sum

impulse response $h(n) = \{1, \underset{\uparrow}{2}, 1, -1\}$

input signal $x(n) = \{\underset{\uparrow}{1}, 2, 3, 1\}$

please find $y(0) = ?$ $\rightarrow$

$$y(n) = \sum_{k=-\infty}^{+\infty} x(k)\, h(n-k)$$

$v_0(k) = x(k) h$

$y(0) \rightarrow \boxed{n=0}$

$$y(0) = \sum_{k=\infty}^{+\infty} x(k)\, h(0-k) = \sum_{k=-\infty}^{\infty} \underbrace{x(k)\, h(-k)} = \sum_{k=-\infty}^{+\infty} v_0(k)$$

$y(0) \rightarrow \boxed{n=0}$

$$y(0) = \sum_{k=-\infty}^{+\infty} x(k)\,h(0-k) = \sum_{k=-\infty}^{\infty} x(k)\,h(-k) = \sum_{k=-\infty}^{+\infty} v_0(k)$$

$$= \sum_{k=-\infty}^{+\infty} v_0(k) = \sum_{k=0}^{3} v_0(k)$$

$$= \underbrace{x(0)}_{1}\,\underbrace{h(-0)}_{2} + \overbrace{x(1)}^{2}\,\overbrace{h(-1)}^{1} + \overbrace{x(2)}^{3}\,\overbrace{h(-2)}^{0} + \underbrace{x(3)}_{1}\,\underbrace{h(-3)}_{0}$$

$$= 2 + 2 + 0 + 0 = \underline{4}$$

Homework Ex.

$$h(n) = \begin{cases} (1/2)^n, & 0 \le n \le 4 \\ 0, & \text{elsewhere} \end{cases}$$

Find the input $x(n)$ for $0 \le n \le 8$ that will generat the output

$$y(n) = \{ \underset{\uparrow}{1}, 2, 2.5, 3, 3, 3, 2, 1, 0, \dots \}$$